MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

Minor Project Proposal :

Design and Implementation of a CHIP-8 System on FPGA with VGA Display

Submitted to : ______________________________

BY: ____________________ [_______________] & ____________________ [_______________]

SECTION: __________

Topic - Design and Implementation of a CHIP-8 System on FPGA with VGA Display

Abstract

This project proposes the design and implementation of a CHIP-8 virtual machine entirely in hardware using Field Programmable Gate Array (FPGA) technology. The system will execute CHIP-8 game ROMs through a Verilog-based processor, store programs and data in 4 KB memory, render the 64×32 monochrome framebuffer on a VGA monitor, and accept user input from a PS/2 keyboard. Delay and sound timers, configurable instruction pacing, and buzzer output will reproduce the essential behavior of the original platform. The work follows a complete digital design flow comprising RTL development, functional simulation, synthesis, FPGA programming, and hardware validation on the DE0-Nano development board.

1. Introduction

FPGA devices provide a flexible platform for studying processor architecture, memory organization, video timing, peripheral interfaces, and real-time digital control. CHIP-8 is a compact interpreted system with two-byte instructions, sixteen general-purpose registers, a 4 KB address space, a 16-level stack, timers, keypad input, and sprite-based graphics. Its small but complete architecture makes it well suited to an educational hardware implementation. This project focuses on building a modular CHIP-8 system in Verilog and displaying its output through standard VGA, thereby connecting digital design concepts with an interactive and visually verifiable application.

2. Problem Statement

Most CHIP-8 emulators run as software on a general-purpose computer, where instruction decoding, graphics, and input handling are performed sequentially by an existing processor and operating system. The problem addressed in this project is to realize the CHIP-8 execution environment directly as synchronous FPGA hardware. The design must correctly coordinate opcode fetch and execution, memory access, sprite drawing and collision detection, 60 Hz timers, keyboard input, VGA timing, and audio output while operating reliably within the timing and resource constraints of the target board.

3. Objectives

- To design and implement a CHIP-8 processor in synthesizable Verilog
- To support the standard CHIP-8 instruction set, registers, stack, memory, and timers

- To generate VGA output from a 64×32 framebuffer scaled for a 640×480 display
- To interface the sixteen-key CHIP-8 keypad through a PS/2 keyboard
- To provide sound-timer output through a passive buzzer
- To simulate, synthesize, program, and validate the complete system on FPGA hardware

4. Proposed System

The proposed CHIP-8 system will consist of the following major modules:

CHIP-8 Processing Unit – Fetches and decodes two-byte opcodes and controls sixteen 8-bit registers, the index register, program counter, stack, delay timer, sound timer, arithmetic operations, branching, and sprite instructions through an FSM-based datapath.

Memory Unit – Provides a 4 KB address space, loads a converted CHIP-8 ROM at the standard address 0x200, and supports runtime data reads and writes.

Display Unit – Maintains the 64×32 monochrome framebuffer, performs XOR sprite drawing and collision detection, and scales each CHIP-8 pixel to a 10×10 VGA pixel block.

Input and Audio Unit – Receives PS/2 scan codes, maps keyboard keys to the hexadecimal CHIP-8 keypad, and drives a 1 kHz buzzer while the sound timer is active.

Timing and Control Unit – Derives VGA pixel enables and configurable instruction-rate pulses from the 50 MHz board clock while keeping the design in a single clock domain.

The modular architecture permits independent verification, simplifies debugging, and supports future extensions such as alternative ROM-loading methods or additional compatibility modes.

5. Methodology

The project will follow a structured development process:

1. Study the CHIP-8 architecture and define functional and timing requirements
2. Develop the CPU, memory, framebuffer, VGA, PS/2, timing, and audio modules in Verilog
3. Convert CHIP-8 ROM files into hexadecimal FPGA memory initialization data
4. Perform module-level and integrated functional simulation using testbenches
5. Synthesize the design in Quartus Prime and program the DE0-Nano board
6. Validate instruction execution, display, keyboard controls, timers, and buzzer using CHIP-8 programs

6. Tools and Technologies

Hardware:

- Terasic DE0-Nano FPGA Development Board (Intel Cyclone IV)
- VGA display and interface circuitry
- PS/2 keyboard and passive piezo buzzer

Software:

- Intel Quartus Prime FPGA Design Software
- Verilog HDL and HDL simulation tools
- Python ROM-to-hexadecimal conversion utility

7. Applications

- Educational demonstration of processor architecture and RTL design
- Practical study of finite state machines, memory systems, and hardware timing
- Demonstration of VGA generation, PS/2 interfacing, and real-time graphics on FPGA
- Foundation for retro-computing systems, soft processors, and hardware emulation projects

8. Conclusion

This project aims to demonstrate a complete interactive computing system implemented directly on FPGA hardware. By combining a Verilog CHIP-8 processor with memory, framebuffer graphics, VGA output, PS/2 keyboard input, timers, and buzzer control, the work provides practical experience across the complete digital system design flow. Successful implementation on the DE0-Nano will verify the feasibility of running CHIP-8 programs without a conventional software processor and will provide a compact platform for further study of computer architecture and FPGA-based system design.